

Cyber-Resilient AI for Robust Autonomous Vehicle Navigation

A Simulation-Based Investigation of Robust Perception, Closed-Loop Driving, and Lightweight Cybersecurity

Sabrina Benghename

Independent Technical Research Project

Technical Research Report

Executive Summary

This project investigates how an autonomous vehicle can continue operating when its visual perception deteriorates and when some of its input data can no longer be trusted.

The work began with a classical lane-detection system based on Canny edge detection and the Hough Line Transform. The visual input was then systematically degraded using noise, blur, low contrast, and occlusion in order to identify the limitations of the classical approach.

A lightweight convolutional neural network, **RobustLaneNet V3**, was then developed to predict five lane-boundary positions directly from simulator images. The trained network was integrated into the vehicle's real closed-loop driving pipeline.

Lane perception is learned by the neural network, while obstacle avoidance, lane-change decisions, steering, throttle, and braking are handled by deterministic planning and mathematical control algorithms.

The complete autonomous system was evaluated under five visual conditions: clean imagery, noise, blur, low contrast, and occlusion.

A final defensive layer was then introduced to investigate **cyber-resilience**. It combines HMAC-SHA256 message-integrity verification, anomaly detection, trusted-data fallback, security logging, and a temporary safe mode.

Key outcome: the autonomous vehicle remained on-road during all five final evaluation conditions. The largest degradation occurred under low-contrast imagery, while the security layer successfully rejected controlled examples of untrusted sensor and AI perception data.

Key Results at a Glance

Metric	Result
Visual conditions evaluated	5
Perception error range	1.67–7.21 px
Stable lane-tracking error range	1.04–8.05 px
Average vehicle speed range	7.22–7.43
Off-road frames	0 in every condition
Controlled cyber scenarios	3

Project repository:

github.com/SabrinaBenghename/Robust-AI-Autonomous-Vehicle-Navigation-with-Cyber-Resilience

1. Project Objective

Autonomous-driving systems depend on reliable perception. A model that works on a clean image may behave very differently when the camera input becomes noisy, blurred, poorly contrasted, or partially obstructed.

The project therefore focused on the following question:

How can a simulated autonomous vehicle maintain reliable lane perception and stable closed-loop behavior under adverse visual conditions while preventing untrusted data from directly influencing vehicle control?

The methodology followed a progressive experimental approach:

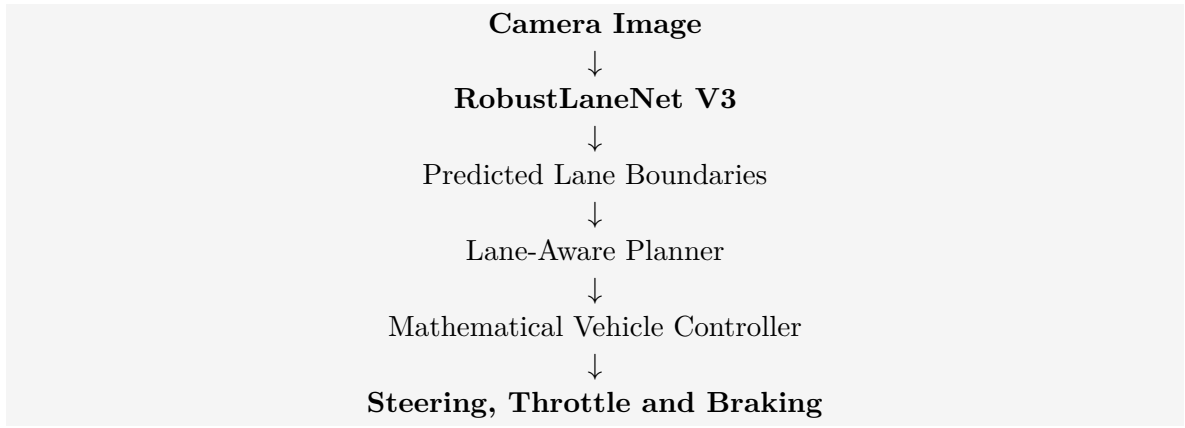
“I established a classical Canny-Hough baseline, systematically degraded the visual input, quantified its failure modes, then investigated a lightweight CNN architecture specifically designed for robustness under adverse visual conditions, and compared the two under identical conditions.”

Instead of evaluating only a neural-network prediction metric, the final stage examined whether perception quality was sufficient for an actual simulated vehicle to remain stable while navigating traffic.

2. Autonomous Driving System

The platform was developed as a custom Python simulation environment combining vehicle dynamics, traffic, camera perception, sensors, planning, control, and experimental robustness testing.

The final pipeline can be summarized as:



The distinction between learned and deterministic components is important.

Component	Method
Lane perception	Trained convolutional neural network
Lane-target estimation	Geometry derived from neural predictions
Obstacle avoidance	Deterministic rule-based planning
Lane-change decisions	Lane gaps, traffic positions and time-to-collision logic
Steering and speed control	Mathematical controller
Cyber anomaly detection	Rule-based validation

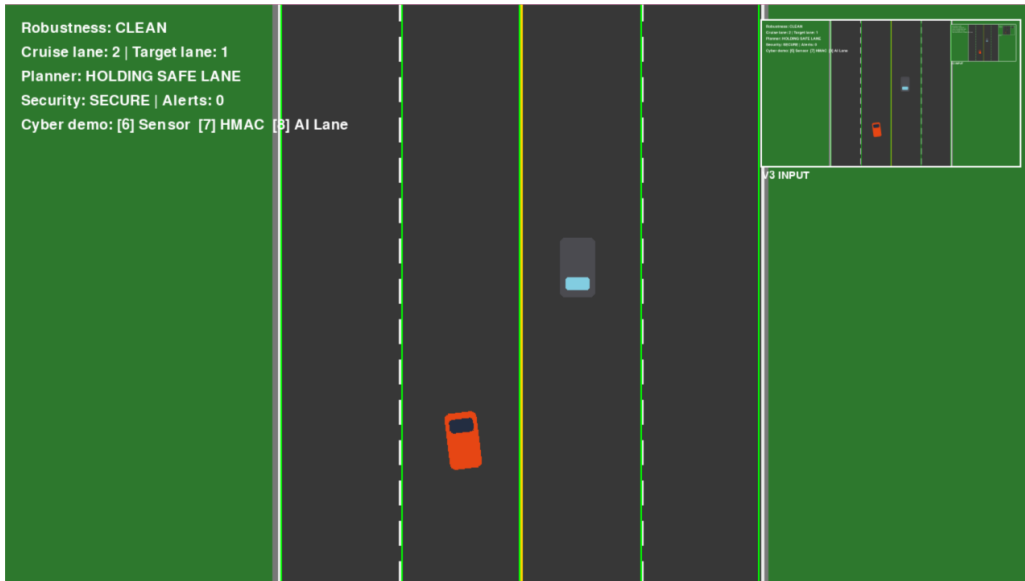


Figure 1: Closed-loop autonomous-driving simulator using RobustLaneNet V3 for lane perception.

3. From Classical Vision to RobustLaneNet V3

3.1 Classical Canny–Hough Baseline

The first perception system used classical computer vision.

The pipeline consisted of:

Image → Grayscale → Canny Edges → Hough Lines → Lane Estimate

This baseline provided an interpretable reference system and allowed visual corruptions to be introduced in a controlled way.

The experiment showed why clean-condition performance alone is insufficient: edge- and line-based assumptions become increasingly fragile as image quality degrades.

3.2 RobustLaneNet V3

The classical experiment motivated the development of **RobustLaneNet V3**, a lightweight convolutional network trained to regress five horizontal lane-boundary locations:

`[road_left, lane_1, center_line, lane_2, road_right]`

A typical prediction is approximately:

`[342, 493, 639, 789, 937]`

The simulator-generated dataset contained 257 labeled images, divided into approximately 179 training samples, 38 validation samples, and 40 test samples.

On the clean test set, RobustLaneNet V3 achieved an average lane-boundary error of approximately **0.68 pixels**. All 40 test images produced predictions within the five-pixel success threshold.

The purpose of the project, however, was not only to obtain low clean-image error. The model was subsequently exposed to the same categories of visual degradation used during the robustness study.

4. Robustness and Closed-Loop Evaluation

The final autonomous-driving experiment evaluated five conditions:

- Clean imagery;
- Gaussian noise;
- Gaussian blur;
- Low contrast;
- Partial occlusion.

The same trained RobustLaneNet V3 model was used in every condition.

Table 1: Final closed-loop autonomous-driving results.

Condition	Coll.	Avoid.	Lane Err.	Percept.	Speed
Clean	3	75%	1.04	2.37	7.31
Noise	3	100%	3.22	1.67	7.26
Blur	3	75%	2.35	4.84	7.35
Low Contrast	3	100%	8.05	7.21	7.22
Occlusion	2	100%	3.28	2.67	7.43

Lane and perception errors are expressed in pixels. No off-road frames were recorded in any condition.

The largest degradation occurred under **low contrast**, which produced an average perception error of 7.21 pixels and an average stable lane error of 8.05 pixels.

Blur was the second most challenging condition for direct perception error.

The noise experiment produced a lower measured perception error than the clean closed-loop experiment. This result should not be interpreted as evidence that noise improves the neural network; the values come from a single controlled run and include normal experimental variation.

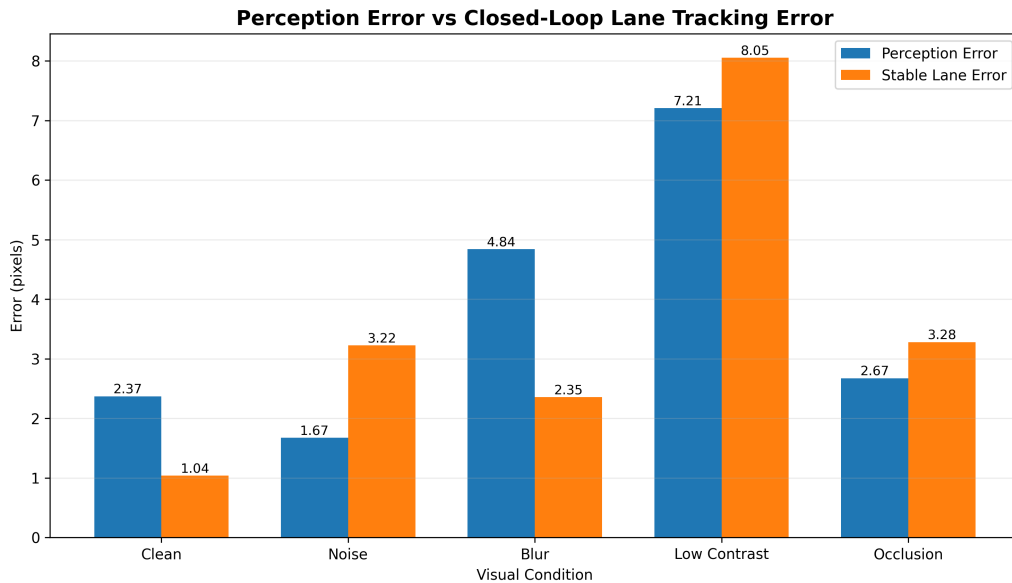


Figure 2: Comparison between neural perception error and closed-loop stable lane-tracking error across the five visual conditions.

An important result is that perception degradation did not automatically lead to loss of lane stability. Despite different levels of visual corruption, **zero off-road frames** were recorded in every final condition.

Average vehicle speed also remained within a narrow interval from 7.22 to 7.43 simulation units.

Collisions were still observed. Because collision counts remained similar across most corruption conditions, they cannot be attributed solely to neural perception degradation. They also expose limitations in the deterministic planning and obstacle-avoidance layer.

5. Lightweight Cyber-Resilience

The final stage investigated a different failure mechanism: data that should not be trusted.

A lightweight defensive security monitor was inserted between perception / sensor acquisition and the planning-control pipeline.

It implements four main protections:

- **HMAC-SHA256 integrity verification** for authenticated sensor messages;
- **plausibility checks** for impossible or suspicious sensor values;
- **validation of neural lane predictions** before they are accepted by the planner;
- **trusted-data fallback and temporary safe mode** when suspicious information is detected.

Three controlled simulator scenarios were implemented.

Sensor Spoofing

A physically impossible front-distance measurement of -500 is injected. The monitor identifies the value as invalid, rejects it, and activates safe mode.

Authenticated Message Tampering

A legitimate sensor message is signed using HMAC-SHA256. The payload is then modified without generating a valid replacement signature.

The integrity check fails and the manipulated message is rejected before it can be trusted by the autonomous-driving pipeline.

AI Lane-Prediction Spoofing

A normal prediction such as

$$[337.4, 488.5, 638.9, 792.5, 940.1]$$

is modified so that one lane boundary becomes approximately 1700 px, which lies outside the image width.

The invalid prediction is rejected and the system temporarily keeps the previous trusted lane estimate.

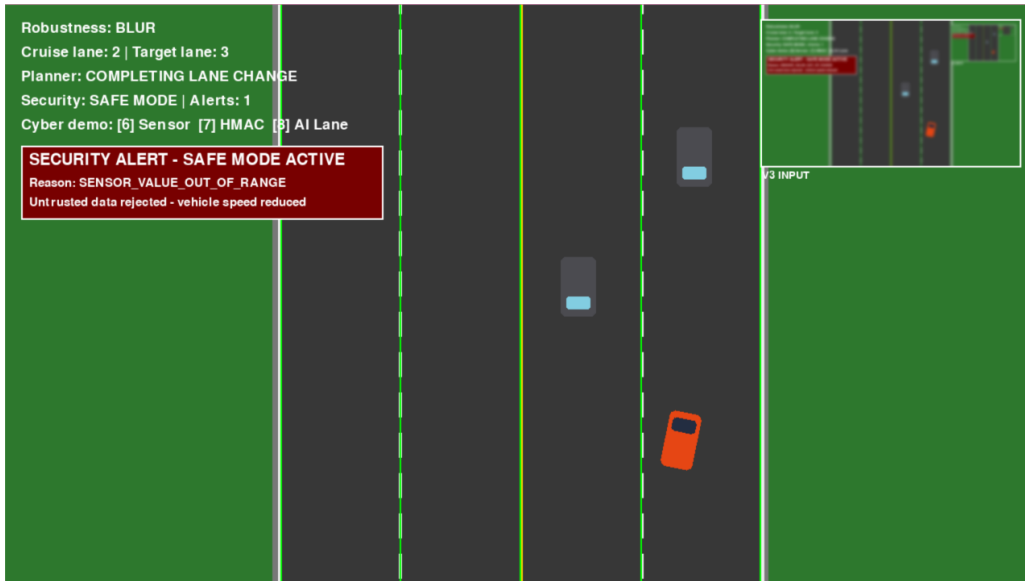


Figure 3: Controlled cyber-resilience demonstration showing security monitoring and temporary safe-mode behavior.

These mechanisms are deliberately lightweight. They demonstrate defensive principles within the simulator and are not intended to represent a complete production automotive intrusion-detection system or real CAN/V2X security architecture.

6. What the Results Mean

The project demonstrates an end-to-end experimental workflow rather than only an isolated machine-learning model.

The most important observations are:

- a neural perception model can be integrated directly into a closed-loop vehicle rather than evaluated only offline;
- low contrast was the most damaging visual condition in the final experiment;
- perception error and actual closed-loop lane error are related but are not identical quantities;
- the vehicle remained on-road under all five tested visual conditions;
- collision performance reveals that robust perception alone does not solve every autonomous-driving problem;
- input validation provides an additional layer of protection when sensor or AI data becomes implausible or untrusted.

The evaluation also has clear limitations. The simulator uses synthetic data and primarily straight road geometry. RobustLaneNet V3 predicts horizontal lane-boundary positions rather than full curved lane polylines. The planner and controller are deterministic, and the final five-condition results correspond to one controlled evaluation episode per condition rather than a large statistical study.

For those reasons, the project should be interpreted as a **research prototype and experimental platform**, not as evidence of production-ready autonomous driving.

7. Conclusion

This project developed a complete simulation-based autonomous-driving research pipeline spanning classical computer vision, deep-learning perception, robustness testing, closed-loop navigation, obstacle avoidance, and defensive cybersecurity.

The work began by establishing and stress-testing a classical Canny–Hough baseline. RobustLaneNet V3 was subsequently developed as a lightweight neural lane-perception model and integrated directly into vehicle control.

Across the final five visual conditions, the vehicle remained on-road while maintaining relatively stable speed. Low contrast produced the strongest degradation, demonstrating that robustness should be evaluated explicitly rather than inferred from clean-image accuracy.

The addition of integrity verification, anomaly detection, trusted-data fallback, and safe-mode behavior further extended the study from *robust perception* toward *robust autonomous systems*.

The resulting platform demonstrates an important engineering principle: autonomous-system reliability depends not on one model alone, but on the interaction between perception, planning, control, robustness, and trust in the data connecting them.

Cyber-Resilient AI for Robust Autonomous Vehicle Navigation

Python · PyTorch · OpenCV · Pygame

[View the complete project and source code on GitHub](#)
